Warehouse of the Future

MIEIC02-DS

Context

- An assembly line for cars that frequently requires a specific list of items from their warehouse.
- Getting an item wrong results in millions in losses.
- Automatically getting the items is too expensive.
- Manually getting them causes a lot of mistakes.

Solution

Add devices to each warehouse container

Create a software that:

- Represents the warehouse
- Displays devices communicating with each other
- Displays devices and their issues
- Proposes new shelf layouts

VERTICALS

- Data lake
- Mesh communication
- Playground
- Layout Optimization

1.
DATA LAKE

Data collection and processing for the warehouse

TEAM PRESENTATION



André Gomes
PRODUCT OWNER



Henrique Pereira
SCRUM MASTER



Afonso Sousa
DEVELOPER



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DEVELOPER



Luís Recharte
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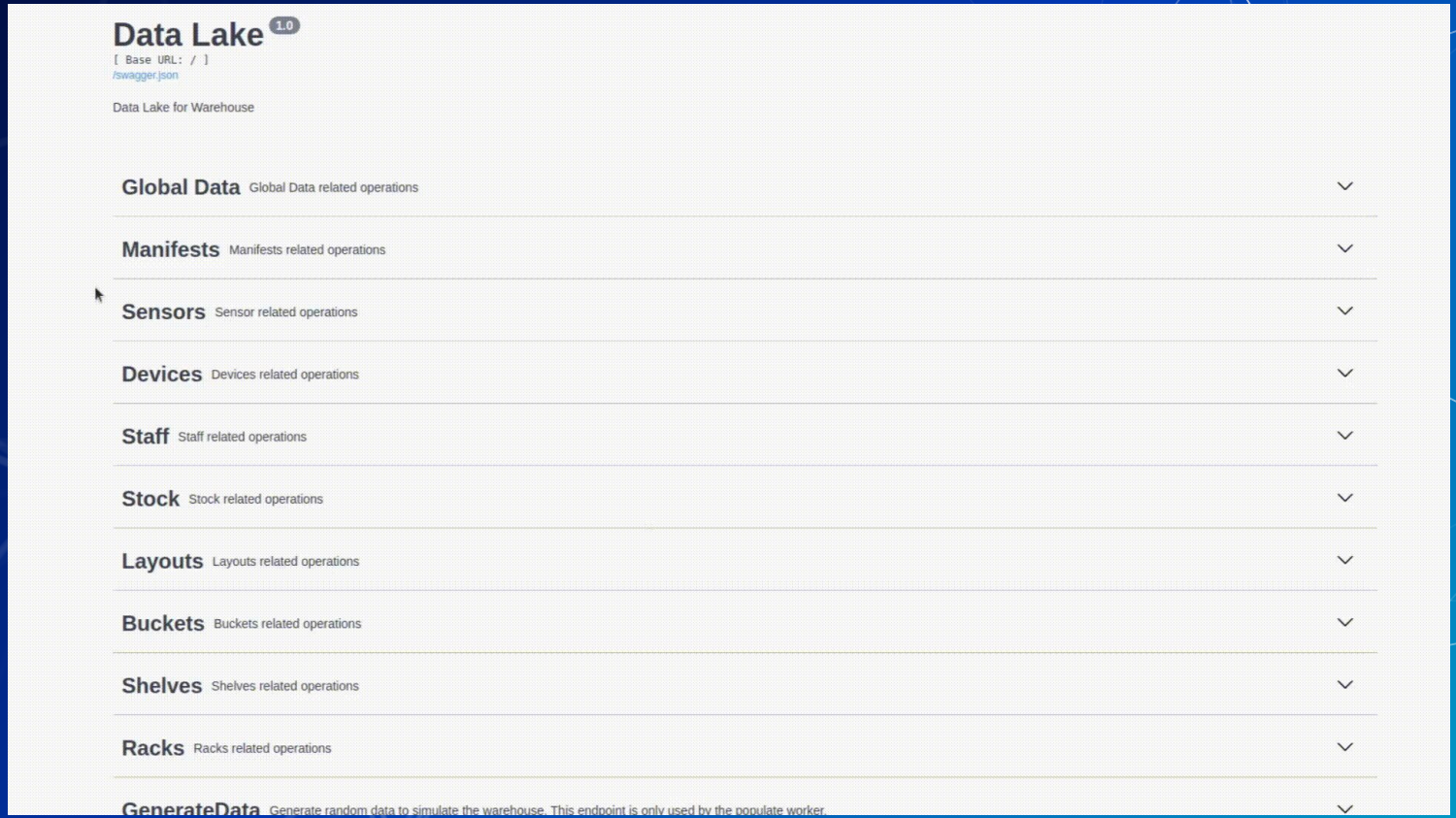


Rui Pinto
DEVELOPER

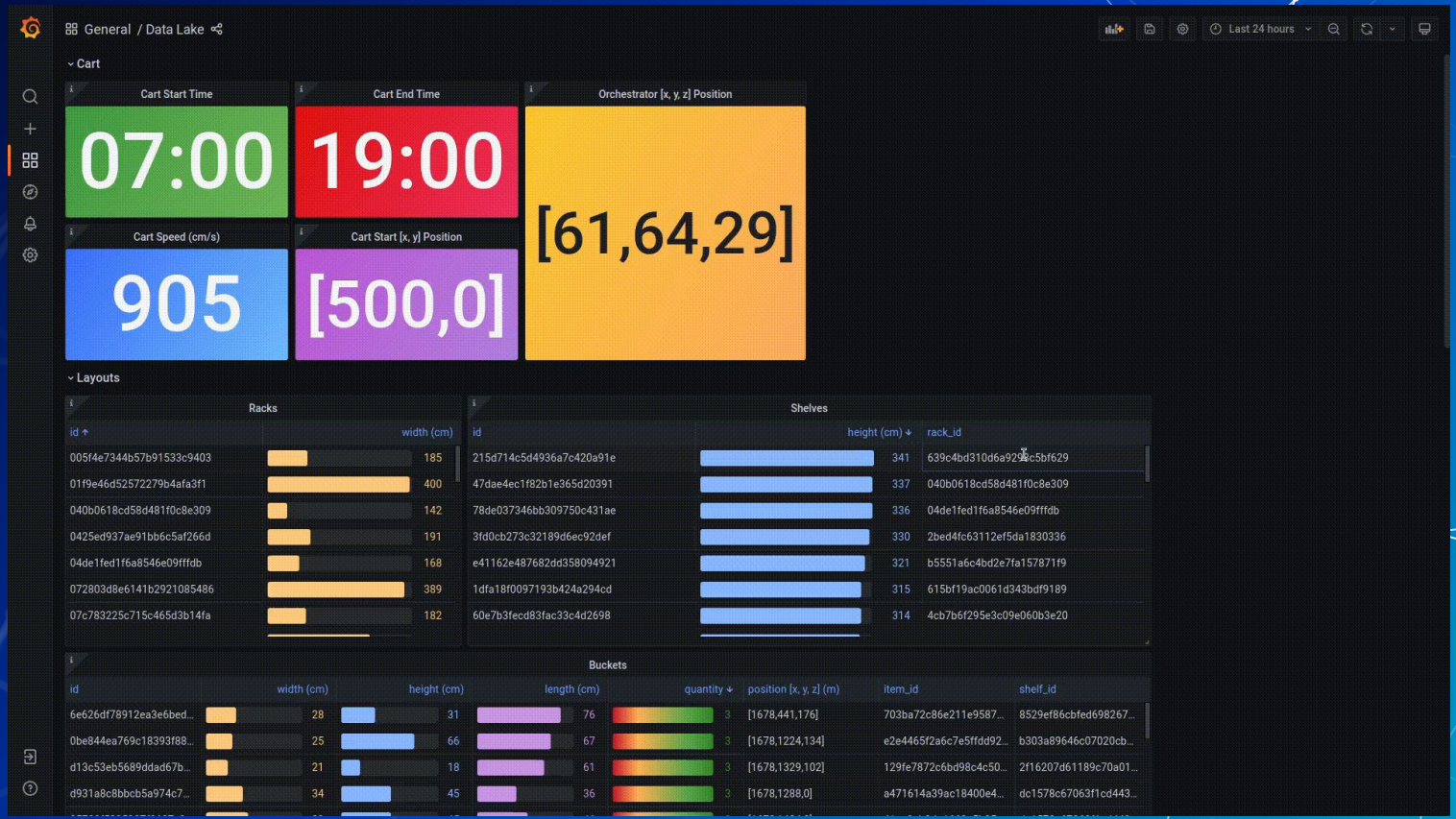


Tiago Gomes
DEVELOPER

REST API


Data Lake 1.0
[Base URL: /]
/swagger.json
Data Lake for Warehouse
Global Data Global Data related operations
Manifests Manifests related operations
Sensors Sensor related operations
Devices Devices related operations
Staff Staff related operations
Stock Stock related operations
Layouts Layouts related operations
Buckets Buckets related operations
Shelves Shelves related operations
Racks Racks related operations
GenerateData Generate random data to simulate the warehouse. This endpoint is only used by the populate worker.

Grafana


General / Data Lake
Last 24 hours
Cart
Cart Start Time
07:00
Cart End Time
19:00
Orchestrator [x, y, z] Position
[61,64,29]
Cart Speed (cm/s)
905
Cart Start [x, y] Position
[500,0]
Layouts
Racks
id
width (cm)
005f4e7344b57b91533c9403 185
01f9e46d52572279b4afa3f1 400
040b0618cd58d481f0c8e309 142
0425ed937ae91bb6c5af266d 191
04de1fed1f6a8546e09fffdb 168
072803d8e6141b2921085486 389
07c783225c715c465d3b14fa 182
Shelves
id
height (cm)
rack_id
215d714c5d4936a7c420a91e 341 639c4bd310d6a929
47dae4ec1f82b1e365d20391 337 040b0618cd58d481f0c8e309
78de037346bb309750c431ae 336 04de1fed1f6a8546e09fffdb
3fd0cb273c32189d6ec92def 330 2bed4fc63112ef5da1830336
e41162e487682dd358094921 321 b5551a6c4bd2e7fa157871f9
1dfa18f0097193b424a294cd 315 615bf19ac0061d343bdf9189
60e7b3fecd83fac33c4d2698 314 4cb7b6f295e3c09e060b3e20
Buckets
id
width (cm)
height (cm)
length (cm)
quantity
position [x, y, z] (m)
item_id
shelf_id
6e626df78912ea3e6bed... 28 31 76 3 [1678,441,176] 703ba72c86e211e9587... 8529ef86cbfed698267...
0be844ea769c18393f88... 25 66 67 3 [1678,1224,134] e2e4465f2a6c7e5ffdd92... b303a89646c07020cb...
d13c53eb5689ddad67b... 21 18 61 3 [1678,1329,102] 129fe7872c6bd98c4c50... 2f16207d61189c70a01...
d931a8c8bbcb5a974c7... 34 45 36 3 [1678,1288,0] a471614a39ac18400e4... dc1578c67063f1cd443...

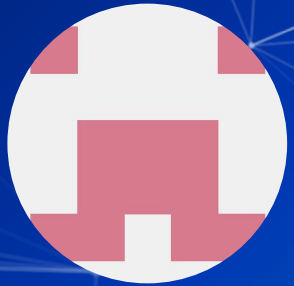
2.
MESH COMMUNICATION

Simulation of the devices' mesh network (visual and text): communication, battery depletion, failure detection, and detailed temporal analysis.

TEAM PRESENTATION



João Costa
PRODUCT OWNER



João Martins
SCRUM MASTER



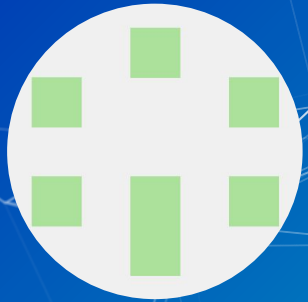
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Davide Castro
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Diogo Rosário
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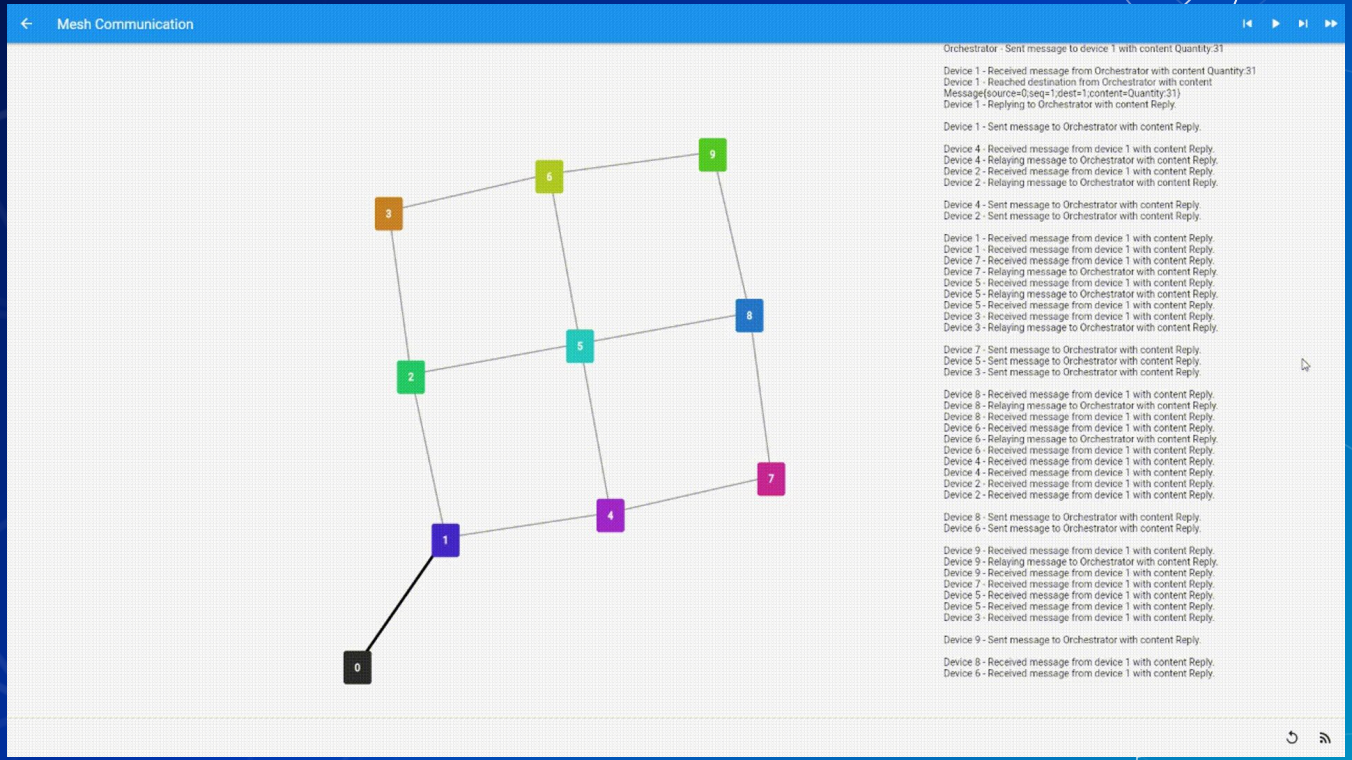


Gustavo Sena
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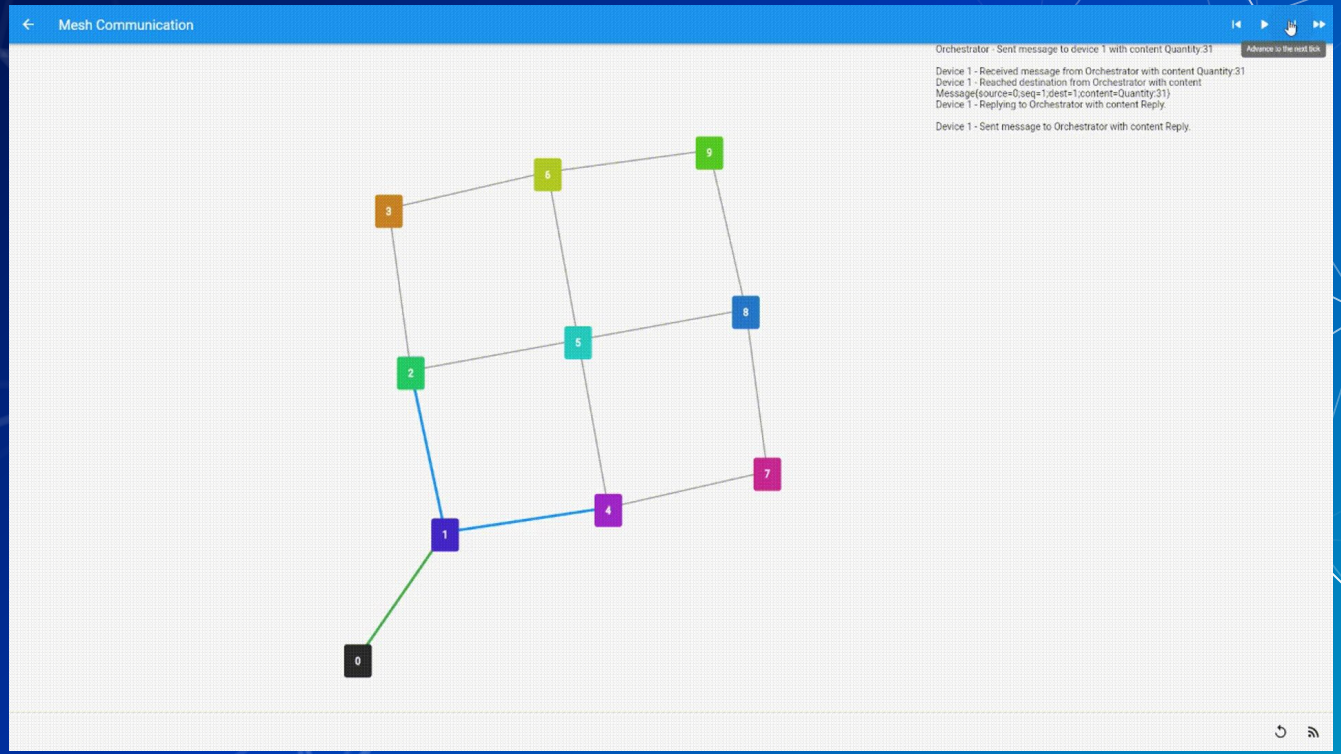


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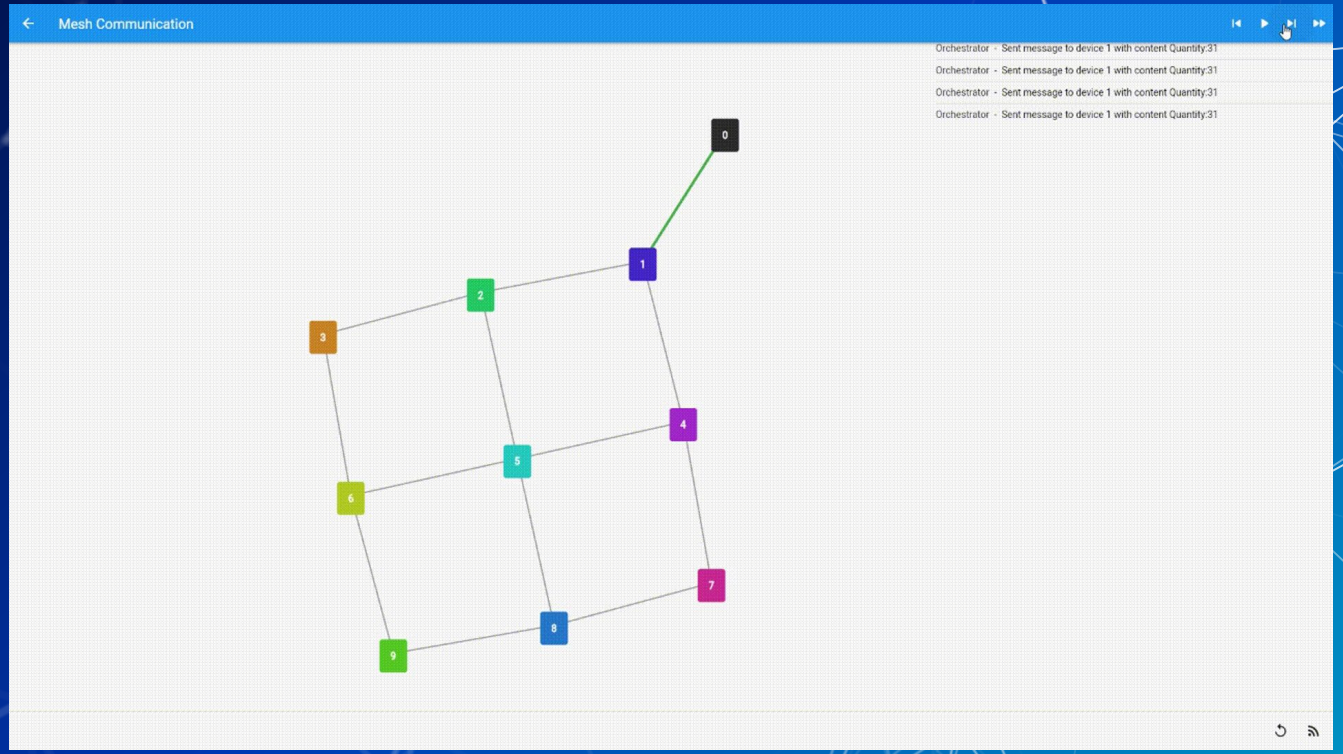
Request and reply



Message relaying



Message failure



User control and logging

The user has several commands available to control the simulation:

- Play/Pause the simulation.
- Step forward in time (single and multi-step)
- Go backwards in time (single and multi-step)
- Command the orchestrator to communicate the current manifest to the devices.

All these actions appear in the logs, so the user can analyze the results in detail.

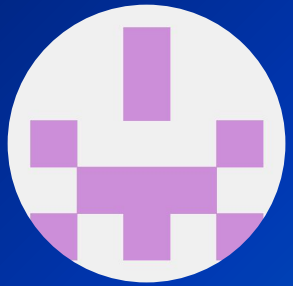
3.
PLAYGROUND

A web-app focused on providing maintenance staff a tool to visualize the real-time status of the warehouse

TEAM PRESENTATION



Tiago Silva
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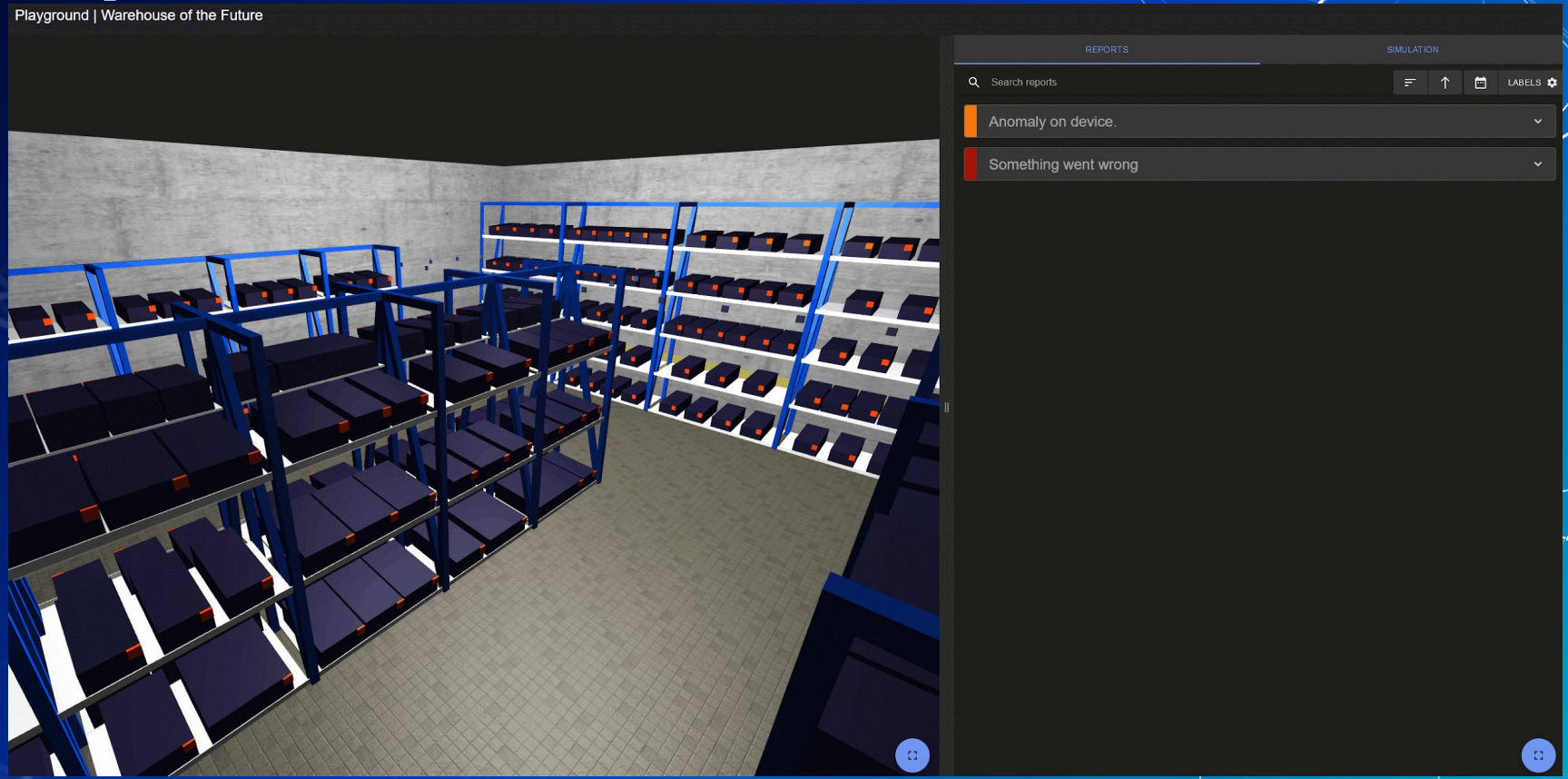


Eduardo Correia
DEVELOPER



Ivo Saavedra
DEVELOPER

Reports


Playground | Warehouse of the Future
REPORTS
SIMULATION
Search reports
LABELS
Anomaly on device.
Something went wrong

Messaging



4.
LAYOUT OPTIMIZATION

Genetic algorithm approach for efficient multi criteria shelf layout

TEAM PRESENTATION



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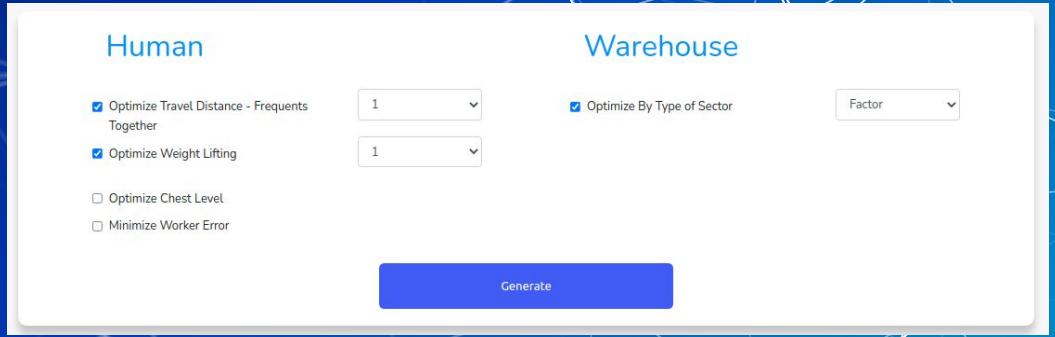
1 200

Average hours wasted each year on unnecessary motion at the warehouse.

U.S. Census Bureau Data

Optimization Criteria

- Weight
- Time
- Human Error Minimization
- Multi Criteria Optimization



Algorithm Result

- Smart Warehouse Organization Layout.
- 2D Results Rendering.
- Connection with Data Lake:
  - Genetic Algorithm Data Feeding



THANK you!

Any questions?

You can find the final project at

- https://github.com/softeng-feup/ds-meic2